

Turn this assignment in on gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

1. Represent $8/9$ as a sum of fractions using Greek Notation (you can use "our" alphabet if you want).

Since the Greeks used unit fractions like the Egyptians, we start with:

$$\begin{aligned}\frac{8}{9} &= \frac{1}{2} + \frac{7}{18} \\ &= \frac{1}{2} + \frac{1}{3} + \frac{1}{18}\end{aligned}$$

Using the notation listed in the book, we have:

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2. Show that the N th triangular number is $\frac{N(N+1)}{2}$ and thus an oblong number is double a triangular number.

The N th triangular number is the sum of the first N natural numbers:

$$T_N = 1 + 2 + 3 + \cdots + N$$

Since this is a finite sum, we can apply the commutative property of addition and rearrange the terms:

$$T_N = N + (N - 1) + (N - 2) + \cdots + 1$$

Adding the two expressions for T_N together, we get:

$$\begin{aligned} 2T_N &= (1 + N) + (2 + (N - 1)) + (3 + (N - 2)) + \cdots + (N + 1) \\ &= (N + 1) + (N + 1) + (N + 1) + \cdots + (N + 1) \\ &= N(N + 1) \end{aligned}$$

Since this is double the triangular number, we divide both sides by 2 to get:

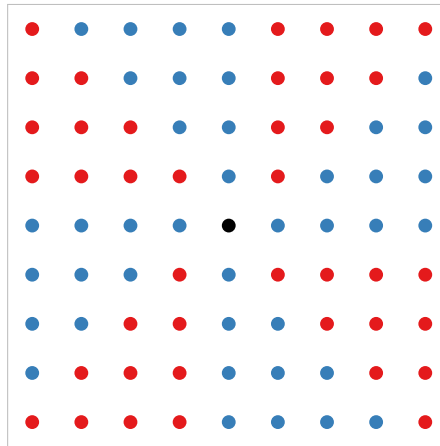
$$T_N = \frac{N(N + 1)}{2}$$

3. Using dots/pebbles, show that eight times any triangular number plus 1 makes a square.

Since the N th triangular number is $\frac{N(N+1)}{2}$, eight times the N th triangular number plus 1 is:

$$8 \cdot \frac{N(N+1)}{2} + 1 = 4N(N+1) + 1 = 4N^2 + 4N + 1 = (2N+1)^2$$

Thus, I know that the square I need to draw will have side length $2N+1$. Here is the diagram for $N=4$:



4. Show that in a Pythagorean triple, if one of the terms is odd, then two of them must be odd and one even.

Let the Pythagorean triple be (a, b, c) such that $a^2 + b^2 = c^2$. We will consider the parity of a , b , and c . Assume that one of the terms is odd. We will analyze the possible cases:

- Case 1: a is odd.
 - If b is odd, then $a^2 + b^2$ is even (since odd + odd = even), making c^2 even, and thus c is even.
 - If b is even, then $a^2 + b^2$ is odd (since odd + even = odd), making c^2 odd, and thus c is odd.
- Case 2: b is odd.
 - If a is odd, then as in Case 1, c is even.
 - If a is even, then as in Case 1, c is odd.
- Case 3: c is odd.
 - If both a and b are even, then $a^2 + b^2$ is even, making c^2 even, contradicting the assumption that c is odd.
 - If one of a or b is odd, this forces the other to be even (since odd + even = odd).

From the above cases, we see that if one of the terms in a Pythagorean triple is odd, then two of them must be odd and one must be even.